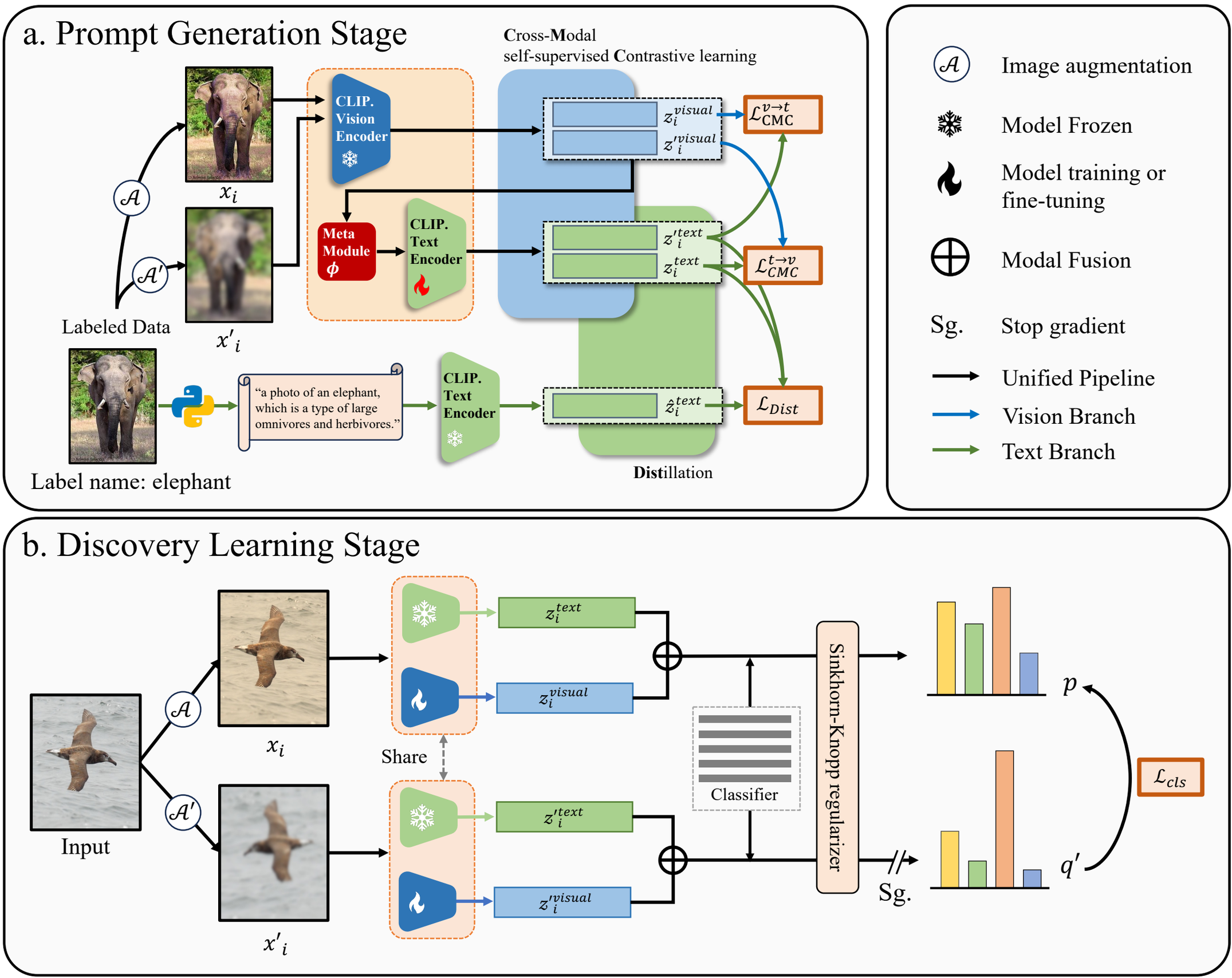
a. Prompt Generation Stage
Cross-Modal
self-supervised Contrastive learning
CLIP. Vision Encoder
Meta Module ϕ
CLIP. Text Encoder
z_i^{visual}
$z_i'^{visual}$
$z_i'^{text}$
z_i^{text}
$\mathcal{L}_{CMC}^{v\to t}$
$\mathcal{L}_{CMC}^{t\to v}$
$\mathcal{A}$
$\mathcal{A}'$
x_i
x'_i
Labeled Data
"a photo of an elephant, which is a type of large omnivores and herbivores."
CLIP. Text Encoder
$\hat{z}_i^{text}$
$\mathcal{L}_{Dist}$
Distillation
Label name: elephant
Image augmentation
Model Frozen
Model training or fine-tuning
Modal Fusion
Sg. Stop gradient
Unified Pipeline
Vision Branch
Text Branch
b. Discovery Learning Stage
Input
Share
Classifier
Sinkhorn-Knopp regularizer
Sg.
p
q'
$\mathcal{L}_{cls}$